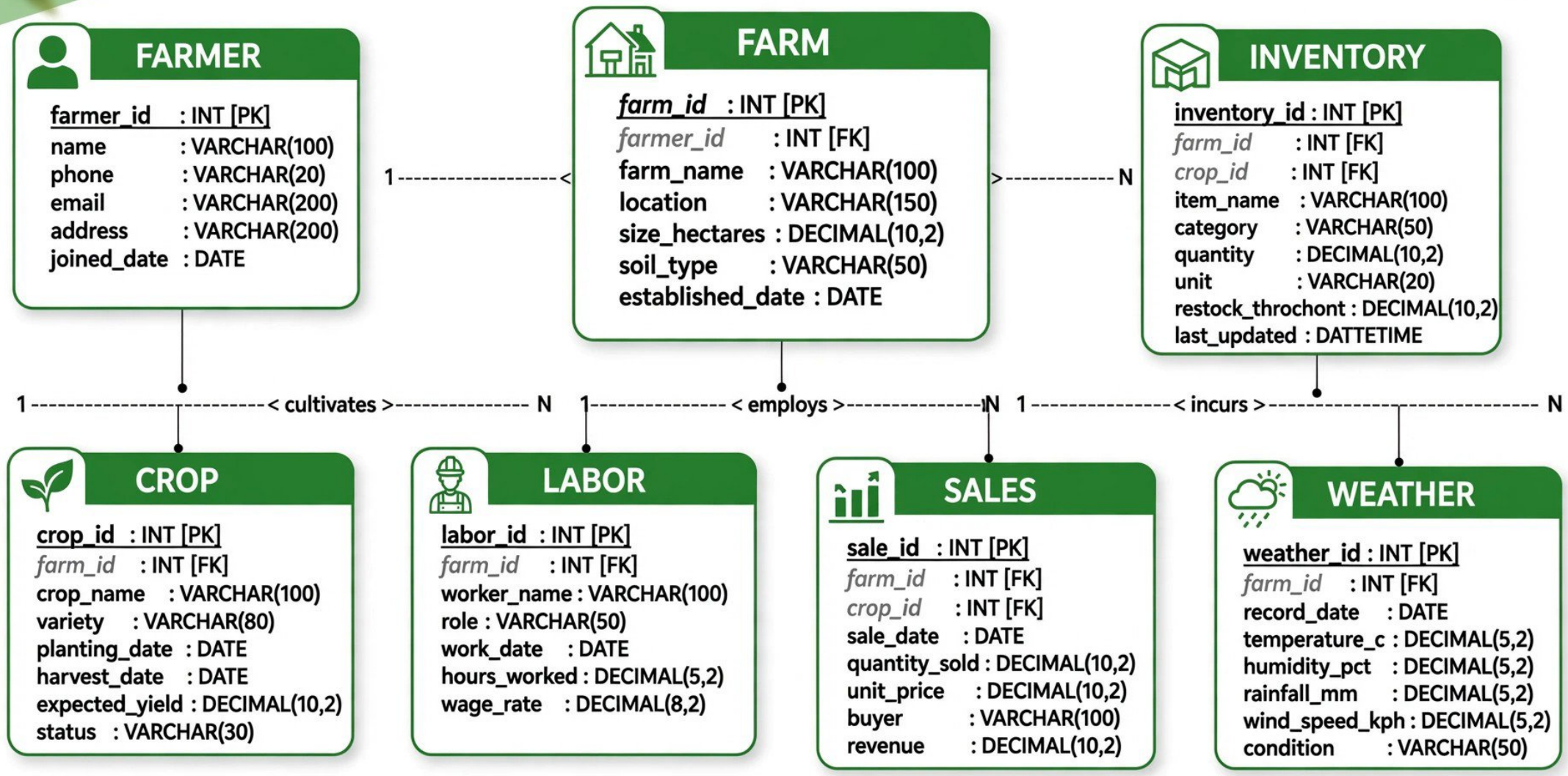
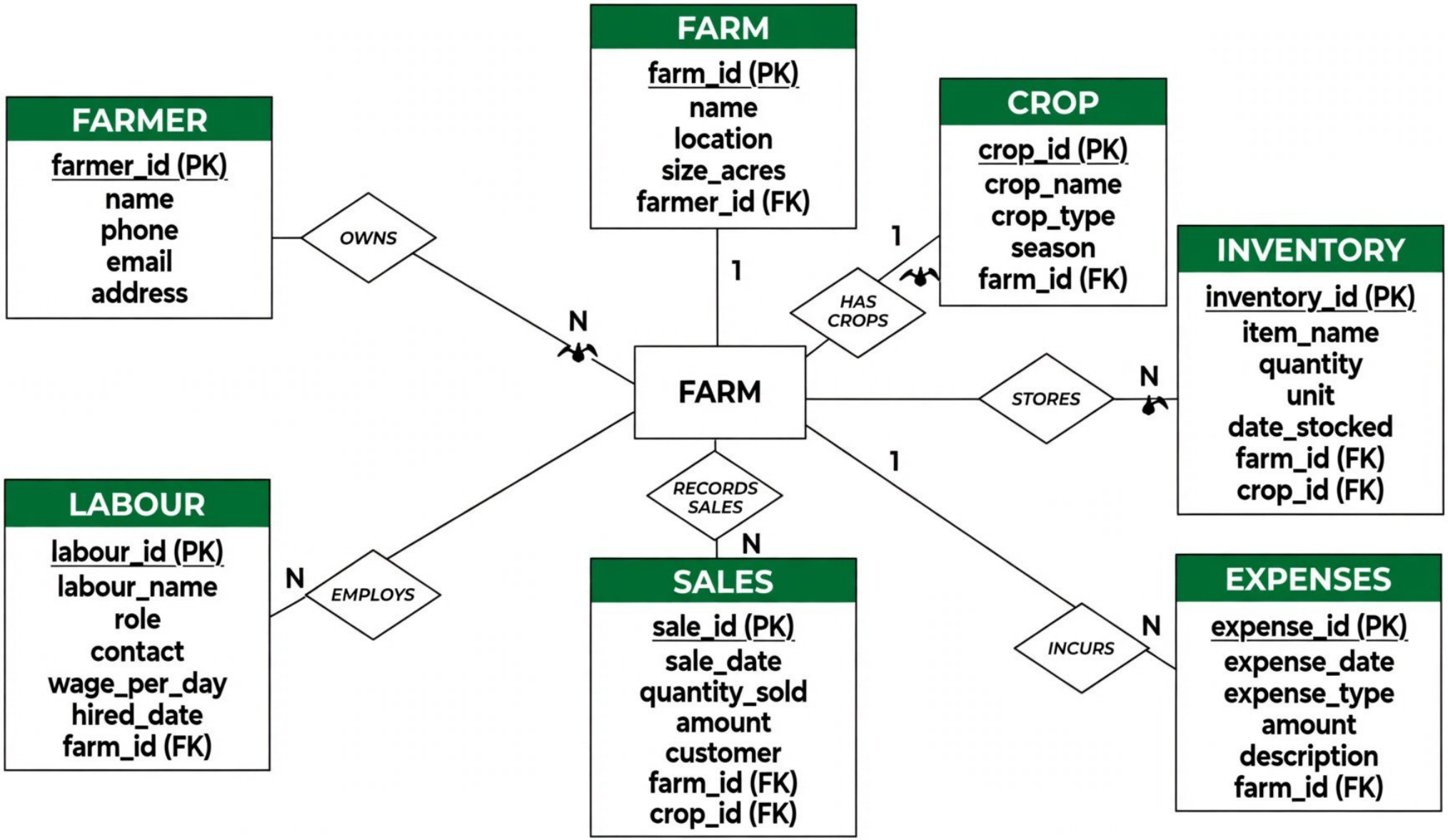
FARM MANAGEMENT SYSTEM - DATA MODEL

Entity-Relationship Diagram • Modern Green Agriculture • PK = Primary Key | FK = Foreign Key | 1:N = One-to-Many Relationship



- Farm is the central entity linking all operational data
- Crop is linked to Inventory & Sales for traceability
- All relationships are 1:N to support scalable records

FARM MANAGEMENT SYSTEM - ER DIAGRAM



Legend: 1 = One • N = Many (Crow's Foot) • andigrared = Primary Key (PK) • (FK) = Foreign Key

FARM MANAGEMENT SYSTEM - DASHBOARD

26 Sept 2026 • Friday

Hi, Alex
Farm Manager

- Overview
- Dashboard
- Crops
- Inventory
- Labour
- Sales
- Expenses
- Weather
- Reports
- Settings
- Help & Support

Good Morning, Alex!

Here's an overview of your farm operations today — 26 Sept 2026

Export Report | + Add New Entry

Crops
12 Active Crops
+2 added this season
↑ Healthy growth

Inventory
340 Items
5 Items Low Stock

Labour
18 Workers
14 Present • 4 On Leave

Sales
$24.5k
↗ +12.4% vs last month

Modules Overview

Manage all key farm operations — live status and insights

Crops
Crops
45.2 ha across 12 Fields
Active Crops: Wheat, Corn, Soybean
Harvest Soon

Inventory
Inventory
340 Items in Storage
Seeds 120kg | Fertilizer 85kg | Pesticide 22L
Restock Needed: 5

Labour
Labour
18 Workers / 14 On Duty
Next Shift: 08:00–16:00 • 4 On Leave
On Schedule

Sales
Sales
$24,500 This Month
42 Orders • Top: Wheat $8.2k | Corn $5.6k
+12.4% ▲ Up

Expenses
Expenses
$9,320 This Month
Fuel $3.2k | Supplies $2.8k | Maintenance $1.9k
-5.2% ▼ Down

Weather
Weather
24°C • Sunny
Humidity 62% | Wind 12 km/h | No Rain
Forecast: Light rain
Partly Cloudy Tomorrow

Reports

View All Reports

- **Crop Yield Report** — Ready
  Generated 25 Sept 2026 • 12.4 tons estimated yield
- **Expenses Summary** — Ready
  Generated 24 Sept 2026 • Total expenses $9,320
- **Labour Timesheet** — Ready
  Generated 24 Sept 2026 • 224 hours logged

Upcoming Activities

- **Harvest Corn**
  Field B12
  Tomorrow • 09:00 AM • Crew Assigned
- **Fertilizer Delivery**
  Today • 02:00 PM • 100kg NPK
- **Labour Weekly Meeting**
  Tomorrow • 10:00 AM • Training & Safety

Soil Moisture Levels

Field A	68%
Optimal • 68%	
Field B	42%
Low • 42% — Irrigation recommended	
Field C	75%
Optimal • 75%	

FARM MANAGEMENT SYSTEM - SECURITY DESIGN

Security Architecture • Defense in Depth • Zero Trust

Layered Security Framework for Agricultural Operations & Data Protection

CORE SECURITY LAYERS

AUDIT LOGS
Comprehensive activity logging • Real-time monitoring • Compliance & forensic analysis

BACKUP
Automated backups • Encrypted backup storage • Disaster recovery & restoration

DATA ENCRYPTION
AES-256 encryption at rest & in transit • Key management (KMS) • Secure data storage

ROLE-BASED ACCESS CONTROL
RBAC policies • Farm Manager / Agronomist / Technician roles • Least privilege access

AUTHORIZATION
Permission management • Access control policies • Resource-level authorization

AUTHENTICATION
Multi-factor authentication (MFA) • Secure login • SSO / OAuth 2.0 support

SECURE API
TLS 1.3 API securiity • Rate limiting & validation • API gateway protection

FIREWALL
Perimeter defense • Intrusion prevention (IDS/IPS) • Network segmentation & monitoring

PRINCIPLE:
Zero Trust Security Model

AGRICULTURE FOCUS:
IoT Sensors, Crop & Livestock Data Protection

COMPLIANCE:
GDPR • ISO 27001 • Data Privacy & Retention

Designed for secure, scalable, and resilient farm management operations

FARM MANAGEMENT SYSTEM - REPORTS DESIGN

Comprehensive Reports Overview | Period: Jan – Dec 2024 | Generated: 2024-12-15

Total Revenue
$85,420
↑ +12.5% vs 2023

Total Expenses
$42,100
↑ +8.2% vs 2023

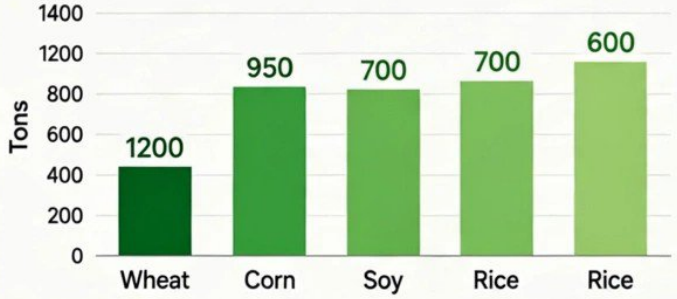
Net Profit
$43,320
↑ +16.8% vs 2023

Crop Yield Avg
3.45 Tons
↑ +5.1% vs 2023

Active Labor
12 Workers
↑ +2 vs 2023

Crop Yield Report

Yield by Crop (Tons)



Expense Report

Expenses Breakdown



Labor 30% Seeds 30% Fertilizer 25% Equipment 15%

Sales Report

Monthly Sales (USD)



Inventory Report

Current Stock Levels

Crop	Stock (kg)	Status
Wheat	2,500 kg	Sufficient
Corn	1,200 kg	Low Stock
Soy	800 kg	Sufficient
Rice	400 kg	Critical

Labour Report

Labor Hours by Activity

